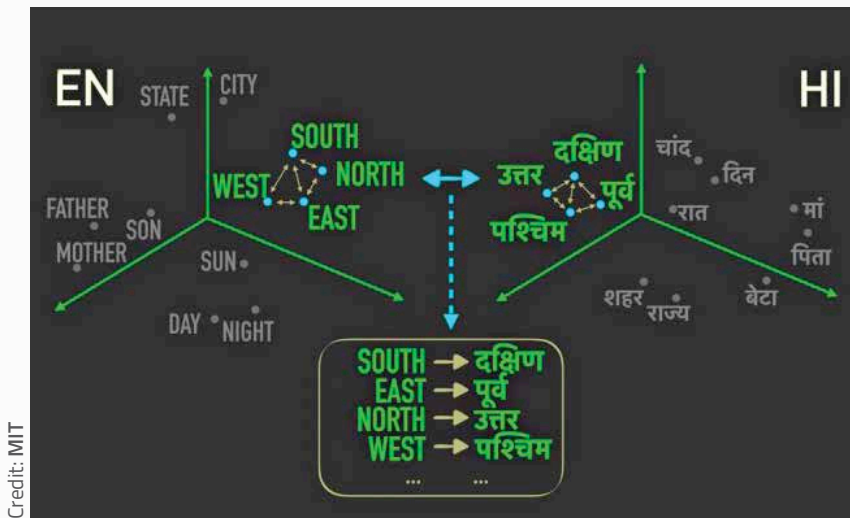


toshop to fix the scratches and blotches on a car, not yet, and certainly not in Hindi. When it comes to training digital assistants, the problem of a dearth in data sets persists. In September last year, Microsoft researchers released free datasets for Telugu, Tamil and Gujarati under the Microsoft Research Open Data initiative. This allowed anyone who wanted to train data for voice-based services in these languages to use the services. Sundar Srinivasan, General Manager of Artificial Intelligence & Research at Micro-

ko call lagana” is more English than Hindi, despite being in Hindi. Acronyms are problematic, even if they are Indian in origin. For example DDLJ is an English acronym made up of Hindi words, but the text string is longer in Devanagari as against English. The ends of sentences may be marked with a full stop or a vertical bar (the poorn viraam). There is a gender agreement between nouns and verbs, which again makes it complicated to navigate and code for. Is the hard disk ‘mehenga’ or ‘mehengi’? Past that, there is the

languages, with support for Hindi, Marathi, Bengali, Tamil, Telugu, Gujarati, Kannada, Malayalam and Urdu. When Assistant received significant updates in 2018, Purvi Shah, Technical Program Manager for Assistant said “The Google Assistant is truly Indian, it’s your helpful dost that speaks our language and understands the things you care about, from finding biryani recipes, to pulling up the latest cricket score, or finding directions to the nearest ATM. In order to make it even more useful to Indians over time, developers and businesses can now build Actions for the Hindi Assistant through the developer platform Actions on Google. Once an action is built, you can just say ‘Ok Google, talk to’ and access the service or content straight through your Google Assistant”. For a comprehensive list of Actions available on the Google assistant for Hindi, head over to <https://dgit.in/hikgg1>. Additionally, Google Assistant allows you to interact with it using two languages, such as English at work and Hindi at home. The bilingual pairing feature works only with Hindi as of now. When it comes to smart speakers however, Google Home only supports Hindi. Right now, Alexa does not support any Indic languages, but does understand some Hinglish commands. This is more of a hit and miss kind of thing, if you actually have an Alexa powered smart speaker. Amazon has a rather unique approach of using a skill called Cleo, to allow users to teach Alexa new languages. The more the skill is used by users, the more Alexa understands local languages. Cleo can be trained in Hindi, Tamil, Marathi, Kannada, Bengali, Telugu and Gujarati. In the back end, developers can add Indic language skills to Alexa. Siri is yet to support Indic languages. However, the iOS 13 update scheduled in September will incorporate Indian male and female accents, as well as better contextual understanding. While Cortana can translate sentences between English and Hindi, Bangla, Tamil, and Urdu, it does not yet natively support Indic languages. 



Credit: MIT

A map of word embeddings in English and Hindi

soft India said, “We believe India’s increasing digital literacy needs to be supported by a multi-lingual digital world. Microsoft Indian Language Speech Corpus is an extension of our on-going efforts to reduce language barriers and empower Indians to harness the full potential of the Internet. Using our technology expertise, we want to accelerate innovation in voice-based computing for India by supporting researchers and academia.”

While companies stand to benefit when they make it as easy as possible for developers to use Indic languages, there are a number of complex, practical considerations. There is a difference between the colloquial and pure form of the languages. Additionally, many terms are borrowed from English by default, with no simple or well-known alternatives in Indic languages. A phrase such as “Housekeeping team

matter of formality embedded in the language. Both Assistant and Alexa treat users with politeness and respect (aap instead of tum) and expect to be spoken to in the same manner. It is not that the other languages are not as or more complex, it is just that these are some of the unique challenges of designing assistants for Indic languages. Every company navigates through these problems in a different way.

For example, Google accepts both Roman and Devanagari script within the code for Actions in Hindi. The word “connect” can be written in either script. However, when creating a new Skill or Action, developers are required to spell out English words in the Roman script only. In both cases, the actual code is a mish mash of the Roman and Devanagari script.

Google Assistant is the leading assistant when it comes to Indic